

Due Date: Week 4

Project ID & Title: #5 — Healthcare Clinic & Telemedicine Portal

A. Project Identity

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Project Title: Centralized Telemedicine Portal and Clinical Management Database System

B. Report Structure

1. Introduction & Project Scope

(Adapted from ISO/IEC/IEEE 29148)

1.0 Problem Statement

Small clinics that still run on paper records and phone calls face a few recurring problems: doctors get double-booked, patient histories are incomplete or scattered across folders, and prescriptions are hard to track. Patients have no way to check a doctor's availability before calling, and staff have no shared record that links an appointment to the diagnosis and prescription that came out of it. This project builds a relational database to fix that — one schema covering patient profiles, doctor schedules, appointment booking, medical records, and prescriptions, with constraints that prevent bad data from entering in the first place.

1.1 System Objective

This project designs a relational database for a small clinic that handles both walk-in visits and remote consultations. The database must store patient profiles, doctor shift schedules, appointment bookings, medical records, and prescriptions in one place — with no duplicate data and no orphaned records. The target is to replace the current paper-and-phone workflow with a schema that a backend application can query directly.

1.1.1 Actor Definitions (Scope & Business Context)

The actors below are defined in the specific context of this project, not as generic roles, to avoid ambiguity in access-control design:

- **Patient (Outpatient only):** An individual who visits the clinic for a scheduled consultation - in-person or telemedicine - and leaves after the appointment. This project models outpatient care exclusively; inpatient admission, ward/bed assignment, and multi-day hospitalization are out of scope, so PATIENT has no admission or discharge attributes.
- **Doctor - Treating Doctor (GP / Specialist):** A clinician directly responsible for a patient's appointment, diagnosis, and prescription. The schema distinguishes doctors by clinical role (GP vs. Specialist, BR-01), since that determines room assignment (BR-03) and specialty data (BR-02). Employment type (Full-time vs. Part-time, see below) is tracked separately and affects compensation only.
- **Doctor - Employment Type (Compensation Basis):** Every doctor is additionally classified as FULL_TIME (fixed monthly salary, independent of shift count) or PART_TIME (compensated per hour actually worked, per SHIFT, at a registered rate). This classification is independent of both the GP/Specialist axis (BR-01) and the Treating/Consulting axis (BR-21 onward) - a doctor can be any combination of the three, e.g. Specialist + Full-time + occasionally

Consulting. Employment type affects compensation only: it does NOT change record-access permissions - both Full-time and Part-time doctors are governed identically by BR-21/BR-22 (access limited to cases where they are the assigned treating doctor, unless invited as a consulting doctor).

- **Doctor - Consulting Doctor (per-case role):** Any registered doctor (GP or Specialist) can additionally act as a Consulting Doctor on a specific case: invited by the treating doctor to review one particular medical record and give a professional opinion, without seeing the patient directly, prescribing, or editing the record. This is not a separate DOCTOR subtype - it is a temporary, per-case relationship represented by the CONSULTATION_REQUEST entity (Section 2.1.2) and governed by BR-21 to BR-23.
- **Admin / Receptionist:** Administrative staff who never provide clinical care. Their scope is limited to doctor account creation, shift/room scheduling infrastructure, the medication catalog, and staff permission management (BR-18 to BR-20); they have no default access to MEDICAL_RECORD or PRESCRIPTION content.

Functional Requirements (Actor-Action-Data Format)

Patient

1. The system must allow a Patient to register and update their demographic and contact profile.
2. The system must allow a Patient to view a Doctor's real-time available time slots before booking.
3. The system must allow a Patient to book an in-person or telemedicine appointment against a confirmed Doctor shift.
4. The system must allow a Patient to view their own appointment history, medical records, and prescriptions.

Doctor

5. The system must allow a Doctor to register their working shifts across designated dates and time slots.
6. The system must allow a Doctor to view and update the status of appointments assigned to them (e.g., CONFIRMED → IN_PROGRESS → COMPLETED).
7. The system must allow a Doctor to record clinical symptoms, diagnosis, and notes for a completed appointment.
8. The system must allow a Doctor to issue a digital prescription containing one or more medication line items for a medical record.

Admin / Receptionist

9. The system must allow an Admin to create and manage Doctor accounts, including specialization type (GP or Specialist).
10. The system must allow an Admin to maintain the clinic's medication catalog (add, update, deactivate items).
11. The system must allow an Admin to assign role-based access permissions to staff accounts.

Non-Functional Requirements (Database Constraints)

- **NFR-01** — Data Integrity: Unique indexes and composite constraints must prevent any two appointments from occupying the same doctor slot at the same time.
- **NFR-02** — Access Control: Role-Based Access Control (RBAC) must be applied at the database level. A patient's medical record is readable only by that patient and the doctors or admin staff who are assigned to their case.

- **NFR-03** — Performance: Common read queries — doctor directory, available time slots, medication catalog — must return results in under 200 ms on a dataset of up to 10,000 patients and 50 active doctors. This will be verified with EXPLAIN ANALYZE in Phase 3.
- **NFR-04** — Security: Columns storing diagnosis text, prescription details, and vital signs must be encrypted at rest (AES-256 at the storage level). Every INSERT, UPDATE, and SELECT against the medical_record table must be written to an audit log table, recording the user, action, and timestamp.

1.2 Business Rules & Constraints

Patient

- **BR-00:** Each patient is uniquely identified in the system (patient_id) and must have at least one verifiable contact method (phone or email) on file.

Doctor Specialization (EER Hierarchy Requirement)

- **BR-01:** Every doctor must be classified as either a General Practitioner (GP) or a Specialist before the account is activated. The specialization is total — a doctor with no subtype cannot be assigned appointments — and disjoint — a doctor cannot hold both subtypes at once. This is enforced at registration; the admin cannot save a doctor record without selecting one type.
- **BR-02:** Each Specialist is associated with a specific medical specialty (e.g., Cardiology, Dermatology, Pediatrics) and must record documented years of practice.
- **BR-03:** Each GP is assigned to a specific consultation room (stored in a ROOM entity) for each shift. No two GPs may be assigned to the same room during overlapping shift times. This constraint will be enforced via a trigger in Phase 3, similar to BR-09.

Doctor Shift Schedules

- **BR-04:** A doctor registers multiple working shift entries. Each shift entry belongs to exactly one doctor.
- **BR-05:** A doctor cannot have overlapping working shift periods on the same calendar date.

Patient Appointments

- **BR-06:** A patient can book multiple appointments, but each appointment belongs to exactly one patient.
- **BR-07:** An appointment is assigned to exactly one doctor.
- **BR-08:** An appointment must be scheduled within an active working shift of the chosen doctor.
- **BR-09:** A doctor cannot have overlapping confirmed appointments (Strict Anti-Double Booking constraint). This rule is enforced at the implementation level via a database trigger or exclusion constraint in Phase 3 — it cannot be expressed as a simple cardinality on the EER diagram.
- **BR-10:** An appointment must be classified as either an in-person clinic visit or a remote telemedicine consultation.
- **BR-11:** The appointment lifecycle status must be constrained to: 'PENDING', 'CONFIRMED', 'IN_PROGRESS', 'COMPLETED', or 'CANCELLED'.

Medical Histories & Records

- **BR-12:** Each APPOINTMENT may produce at most one MEDICAL_RECORD, but only after the appointment status is set to 'COMPLETED'. A MEDICAL_RECORD must always

reference a valid parent appointment (FK NOT NULL); it cannot exist on its own. In the EER diagram, the relationship from APPOINTMENT to MEDICAL_RECORD is (1 : 0..1) — one appointment, zero or one record — because pending or cancelled appointments never generate one.

- **BR-13:** A medical record stores vital signs, symptoms, clinical diagnosis, and notes recorded by the attending doctor.

Digital Prescriptions & Medications

- **BR-14:** A completed medical record can result in at most one digital prescription. A prescription cannot exist without an associated medical record.
- **BR-15:** A digital prescription can contain one or multiple medication line items.
- **BR-16:** A single medication can appear in many prescriptions, and one prescription can list many medications (M:N). This is resolved through an associative entity PRESCRIPTION_ITEM, which links PRESCRIPTION and MEDICATION and carries the per-item details: quantity, dosage, frequency, and duration.
- **BR-17:** Each PRESCRIPTION_ITEM must specify a positive quantity, exact dosage instructions, intake frequency, and treatment duration in days.

Admin & Access Control

- **BR-18:** An ADMIN account must exist in the system before any Doctor account can be created or activated. Each ADMIN is uniquely identified (admin_id) and must hold a designated role (e.g., 'RECEPTIONIST' or 'SYSTEM_ADMIN'). This is enforced by a NOT NULL FK reference from DOCTOR.created_by to ADMIN.admin_id.
- **BR-19:** An Admin may add, update, or logically deactivate (is_active = FALSE) a medication entry in the MEDICATION catalog. A medication record cannot be hard-deleted if it is referenced by any existing PRESCRIPTION_ITEM; deactivation is used instead to preserve historical prescription integrity.
- **BR-20:** Role-Based Access is managed through a STAFF_PERMISSION table linking ADMIN accounts to a set of permitted actions (e.g., 'CREATE_DOCTOR', 'MANAGE_MEDICATION', 'VIEW_ALL_RECORDS'). Each permission assignment records the admin who granted it and the timestamp, supporting the audit requirement in NFR-04. An admin cannot grant permissions exceeding their own role level.

Clinical Access Control & Consultation

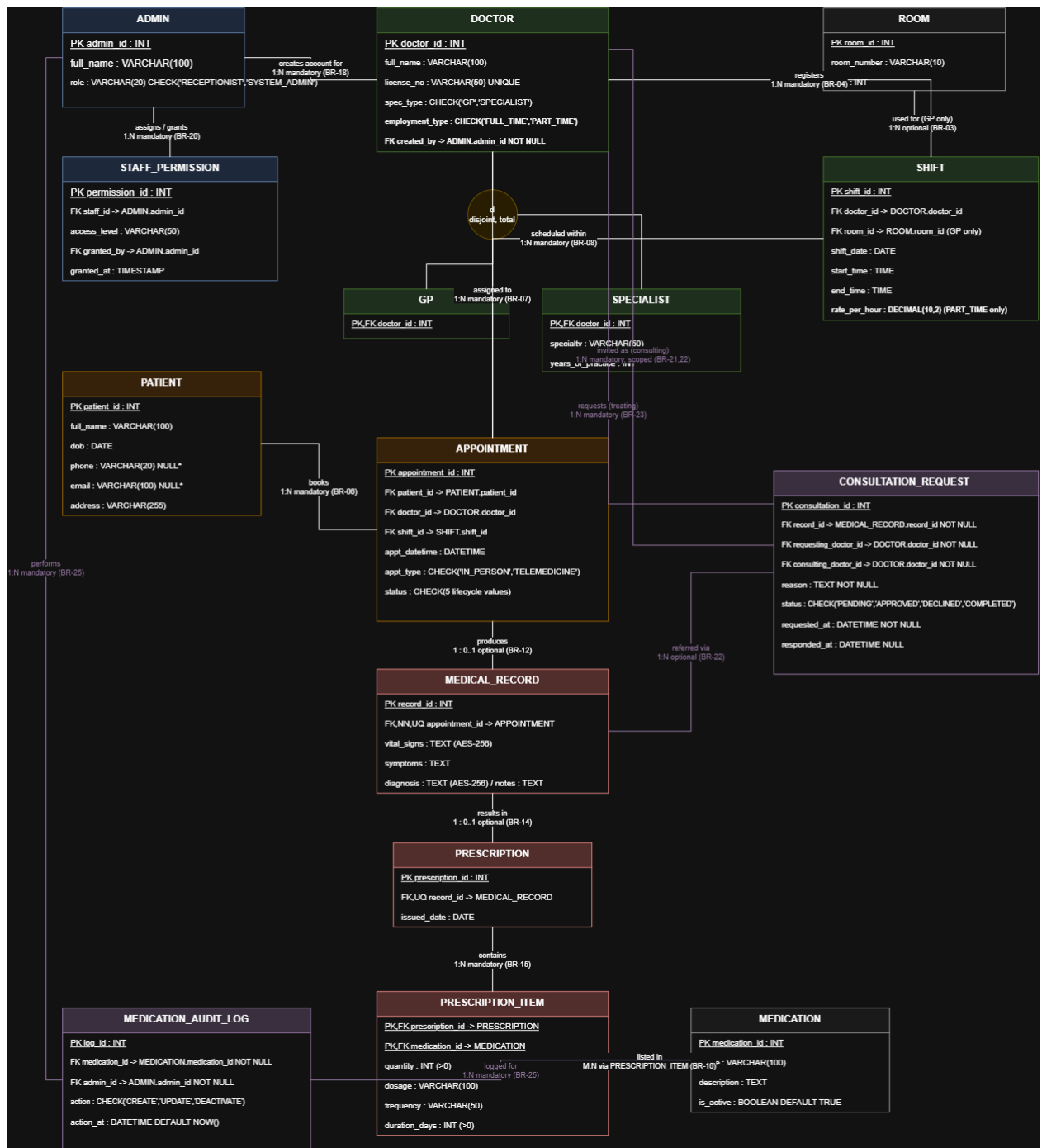
- **BR-21:** By default, a doctor (GP or Specialist) may view the MEDICAL_RECORD and PRESCRIPTION of a patient only for appointments where that doctor is the assigned treating doctor (APPOINTMENT.doctor_id). A doctor cannot browse a patient's records from other doctors or specialties (e.g., a dentist cannot view a cardiology record of the same patient) merely because the patient is shared.
- **BR-22:** Cross-specialty or second-opinion access is granted only through a CONSULTATION_REQUEST: the treating doctor invites a named consulting doctor to review one specific medical record. The consulting doctor's access is read-only, scoped to that single record, and limited to the period the request remains 'APPROVED'; the consulting doctor cannot issue a prescription or edit the record.
- **BR-23:** Each CONSULTATION_REQUEST must record the requesting (treating) doctor, the invited consulting doctor, the target medical record, a reason, and a status ('PENDING', 'APPROVED', 'DECLINED', 'COMPLETED'), satisfying the audit trail required for access to sensitive medical data (NFR-04).
- **BR-24:** Every DOCTOR has an employment_type ('FULL_TIME' or 'PART_TIME'), a mandatory attribute independent of spec_type (BR-01). If employment_type = 'PART_TIME',

every SHIFT belonging to that doctor must have a rate_per_hour (DECIMAL, > 0) to compute compensation; if employment_type = 'FULL_TIME', rate_per_hour is NULL, since salary is fixed and managed outside this system's scope. This rule affects compensation calculation only and does NOT alter BR-21/BR-22 - medical-record access scope is identical for Full-time and Part-time doctors.

- **BR-25:** Every CREATE, UPDATE, or DEACTIVATE action performed on a MEDICATION record must be logged in MEDICATION_AUDIT_LOG, recording the acting admin, the action type, and the timestamp. This resolves the previously descriptive-only "ADMIN manages catalog MEDICATION" relationship on the EER diagram, which had no backing FK or associative entity. Unlike a single managed_by column, this design preserves the full history of every change, consistent with the audit intent of NFR-04.

2. Database Design (ISO/IEC 19505 / IE Standards)

2.1 Conceptual Model (ER/EER Diagram)



2.1.1 Entity Definitions

Each entity in the EER diagram above represents the following business concept:

- **ADMIN:** A staff member with administrative privileges who manages doctor accounts, the medication catalog, and role-based access permissions (BR-18, BR-19, BR-20).
- **STAFF_PERMISSION:** A specific access permission granted to an admin account, recording which action is allowed and by whom (BR-20).
- **DOCTOR:** A clinician who registers shifts and is assigned to appointments. Every doctor is classified as either a GP or a Specialist (BR-01).
- **GP:** Subtype of DOCTOR representing a General Practitioner, who is assigned to a fixed consultation room per shift (BR-03).

- **SPECIALIST:** Subtype of DOCTOR representing a doctor with a declared medical specialty and years of practice (BR-02).
- **ROOM:** A physical consultation room used exclusively for GP shifts (BR-03).
- **SHIFT:** A block of time during which a doctor is available for appointments, on a specific date and time range (BR-04, BR-05).
- **PATIENT:** An individual who registers a profile in order to book appointments and view their own medical history (BR-00).
- **APPOINTMENT:** A booking made by a patient against a doctor's shift, either in-person or via telemedicine, tracked through a defined status lifecycle (BR-06 to BR-11).
- **MEDICAL_RECORD:** The clinical outcome of a completed appointment, capturing vital signs, symptoms, diagnosis, and doctor's notes (BR-12, BR-13).
- **PRESCRIPTION:** A digital prescription issued from a medical record (BR-14).
- **PRESCRIPTION_ITEM:** An associative entity resolving the M:N relationship between PRESCRIPTION and MEDICATION, carrying per-item dosage details (BR-15, BR-16, BR-17).
- **MEDICATION:** An item in the clinic's medication catalog, which can be deactivated but not deleted once referenced by a prescription (BR-19).
- **MEDICATION_AUDIT_LOG:** A full history record of every CREATE, UPDATE, or DEACTIVATE action taken on a MEDICATION entry, recording which admin performed it and when (BR-25).